

Purpose

The purpose of this JHA is to identify and control hazards associated with **RPAS (Remotely Piloted Aircraft Systems)** operations in work environments, including construction, inspection and survey activities. Operating RPAS introduces unique safety risks (collision, uncontrolled descent, equipment failure, and line of sight challenges) which must be systematically controlled to ensure worker safety and regulatory compliance.

Scope

This JHA applies to all personnel involved in **RPAS planning, pre-flight checks, launch, in-flight control, data collection, and recovery** phases, including pilots, spotters, and ground crew. It covers flying near workers, above worksites, near overhead powerlines, and within restricted environments.

PPE Requirements

1. **High-Visibility Clothing** – Required when operating near vehicles/equipment.
2. **Hard Hat** – Required in work areas where debris or drone parts might fall.
3. **Eye Protection** – When performing inspection or maintenance.
4. **Cut-Resistant Gloves** – For battery and equipment handling.
5. **Hearing Protection** – Where rotary noise levels exceed safe thresholds.
6. **Weather-Appropriate Outerwear** – For work in exposed outdoor environments.

Emergency Procedures

1. **Loss of Control / Unplanned Descent:** Immediately shut down motors if safe; ensure landing zone is clear of personnel.
2. **Flyaway / Loss of Signal:** Abort mission and execute return-to-home; if none available, follow emergency procedures and notify site supervisor.
3. **Battery Fire or Thermal Runaway:** Use appropriate fire extinguisher (Class D or lithium fire response), evacuate area, and call emergency services.
4. **Collision with Personnel or Equipment:** Cease operations, provide first aid, secure scene, report incident per site policy.
5. **Electrical Shock During Charging/Equipment Use:** Isolate power, do not touch victim if energized, call emergency services.

Training Requirements

1. **RPAS Pilot Certification:** Completion of approved RPAS pilot training (Transport Canada basic or advanced, depending on operation).
2. **Pre-Flight Inspection & Equipment Familiarization:** Documented training on specific RPAS model and payloads.
3. **Hazard Assessment & Controls:** Site-specific hazard awareness, emergency response procedures.
4. **Emergency Procedures & First Aid:** TRU / First Aid training appropriate for site.

5. **Battery Handling & Fire Response:** Safe charging, storage, and fire containment training (lithium battery safety).
6. **Traffic & Site Awareness:** Workers should understand site traffic patterns, exclusion zones, and work area demarcation.

Job Hazard Analysis Table

Activity	Potential Hazard	Risk Level	Control Measures	Relevant Legislation / Guidance
Pre-Flight Planning & Site Assessment	Poor weather forecast, inadequate risk assessment, powerline proximity	High	- Review weather conditions (wind, rain, visibility). - Conduct site assessment; identify obstacles, restricted airspace, line of sight issues. - Establish flight safety brief for crew.	OHS Code Part 2 – Hazard assessment, control and planning; Transport Canada CARs (RPAS operational restrictions)
Pre-Flight Equipment Inspection	Mechanical/electrical failure, battery malfunction	High	- Perform equipment inspection checklist (airframe, motors, batteries, GNSS signal, controller comms). - Verify firmware/controls are updated. - Use only manufacturer-approved batteries and components.	OHS Code Part 2 – Hazard assessment, control and maintenance obligations
Launching RPAS	Loss of control, fail to clear area, rotor strike	High	- Ensure launch/landing area clear of personnel and overhead obstructions. - Maintain communication with ground crew; use designated spotter where required.	OHS Code Part 2 – Establish controls before work begins

			- Announce takeoff and landing.	
Flying / In-Flight Operations	Mid-air collision, uncontrolled descent, loss of signal	High	- Pilot maintain Visual Line of Sight (VLOS) at all times. - Maintain safe separation from workers, equipment, structures. - Abort flight if unexpected hazards arise or signal is lost.	OHS Code Part 2 – Maintain control of hazards; Transport Canada CARs – RPAS flight rules
Operating Over People or Work Areas	Falling equipment, rotor wash, distraction	High	- Do not fly directly over workers unless essential; set up exclusion zones. - Use signage/barriers; keep spectators at safe distance. - Maintain buffer zones from energized equipment.	OHS Code Part 9 s. 157 (fall hazards) + Part 2 (hazard controls)
Battery Charging / Handling	Fire, explosion, chemical exposure	Medium	- Charge in designated fire-resistant area, away from flammable materials. - Use approved chargers and follow manufacturer protocols. - Monitor charging, do not leave unattended.	OHS Code Part 18 – PPE & general battery safety practices
Data Transfer / Download	Trip hazards (cords), electrical shock	Low	- Keep cords organized; use surge-protected power bars. - Shut down devices properly; avoid water contact.	OHS Code Part 9 – General safeguards
Emergency Recovery (Forced Landing)	Injury from uncontrolled landing, collision	High	- Establish emergency landing procedures before flight. - Identify safe recovery	OHS Code Part 2 – hazard control and

			areas in planning stage. - Abort flight and land immediately when control is compromised.	emergency planning
Post-Flight Inspection & Storage	Burns (hot motors), battery damage, trip hazards	Medium	- Allow equipment cool down before storage. - Inspect for damage; remove compromised components. - Store batteries in approved containers/locations.	OHS Code Part 2 – ongoing hazard control; Part 18 – PPE and storage

Relevant Legislation / Guidance Sources

- **Alberta OHS Code – Part 2:** Hazard assessment and control
- **Alberta OHS Code – Part 9 s. 157:** Fall hazards impacting people under elevated equipment
- **Alberta OHS Code – Part 18:** PPE requirements
- **Transport Canada – CARs (RPAS Regulations):**
 - Standard Operating Rules for RPAS
 - Visual Line of Sight (VLOS) requirements
 - RPAS operator and pilot certification requirements
 - RPAS flight restrictions over people/structures